

Skills Summary

Index-Shift and Convergence-Condition Errors

Use this reference while completing your worksheet or when studying.

Skill 1 — Counting the terms a sum really has

Rule: a sum from p to q has $q - p + 1$ terms, not q of them. Standard formulas such as $1 + 2 + \cdots + n = \frac{n(n+1)}{2}$ start at 1; to start higher, compute the whole thing and subtract the terms you do not want.

Example: Evaluate $\sum_{k=3}^7 k$.

Step 1. The closed formula starts at 1, so build the sum to 7 and remove the part below 3.

Step 2. $\frac{7 \cdot 8}{2} - (1 + 2) = 28 - 3$, and the count checks out: $7 - 3 + 1 = 5$ terms.

$$\Rightarrow \sum_{k=3}^7 k = 25$$

Try it: Evaluate $\sum_{k=4}^8 k$.

check: 80

Skill 2 — Shifting an index moves both limits

Rule: substituting $m = n - c$ replaces n by $m + c$ in the term *and* lowers both limits by c . Check the shift by comparing the first term before and after — they must be the same number, and the term count must not change.

Example: Evaluate $\sum_{n=1}^3 2^n$, then check it against the shifted form $\sum_{m=0}^2 2^{m+1}$.

Step 1. Both forms should list the same three terms, so evaluate the original and use the count as the check on the shift.

Step 2. $2 + 4 + 8$, and the shifted form gives $2^1 + 2^2 + 2^3$ — the same three terms.

$$\Rightarrow \sum_{n=1}^3 2^n = 14$$

Try it: Evaluate $\sum_{n=2}^4 2^n$.

check: 82

Skill 3 — When a geometric series has a sum

Rule: $\sum ar^n$ converges only when $|r| < 1$, and then the sum is $\frac{a}{1-r}$, where a is the series' first term as written. If $|r| \geq 1$ there is no finite sum, and applying the formula anyway can return a nonsense value such as a negative number for a series of positive terms.

Example: Evaluate $\sum_{n=0}^{\infty} \left(\frac{1}{3}\right)^n$.

Step 1. Check the ratio first: $r = 1/3$ has absolute value below 1, so the sum formula is legal here.

Step 2. The first term is the $n = 0$ term, 1, so the sum is $\frac{1}{1 - 1/3} = \frac{1}{2/3}$.

$\Rightarrow$ the sum is $\frac{3}{2}$

Try it: Evaluate $\sum_{n=1}^{\infty} 4\left(\frac{1}{2}\right)^{n-1}$.

check: 8

Skill 4 — Limits of rational sequences

Rule: for a ratio of polynomials in n , divide top and bottom by the highest power of n . Constants then vanish but *coefficients do not*: the limit is the ratio of the leading coefficients, not 1. A sequence whose terms alternate in sign has no limit at all, however well the sizes settle down.

Example: Find $\lim_{n \rightarrow \infty} \frac{5n-1}{2n+3}$.

Step 1. Top and bottom both grow like n , so divide through by n and let the constant pieces die.

Step 2. $\frac{5 - 1/n}{2 + 3/n} \rightarrow \frac{5 - 0}{2 + 0}$.

$\Rightarrow$ the limit is $\frac{5}{2}$

Try it: Find $\lim_{n \rightarrow \infty} \frac{2n+7}{n-4}$.

check: 2